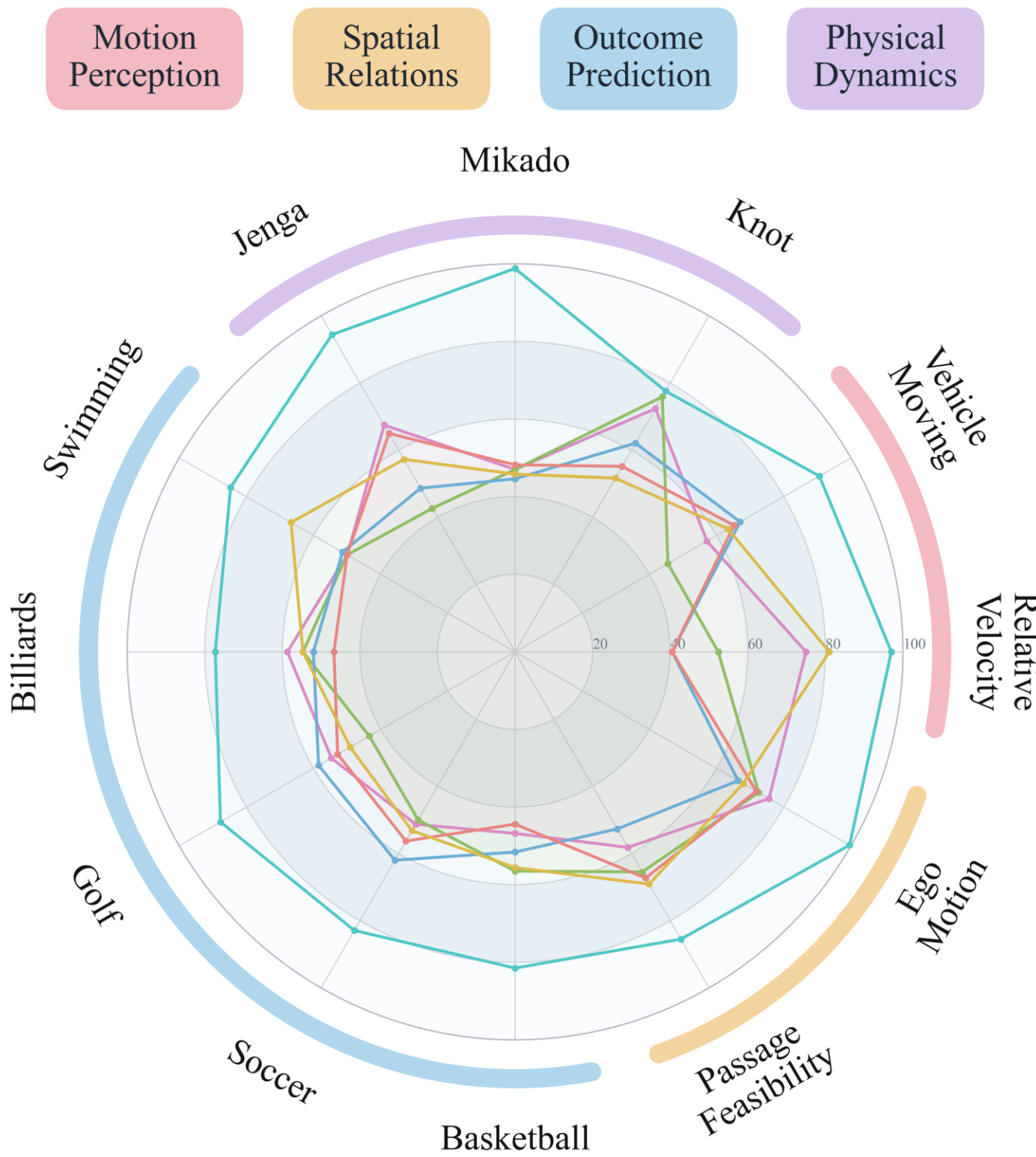
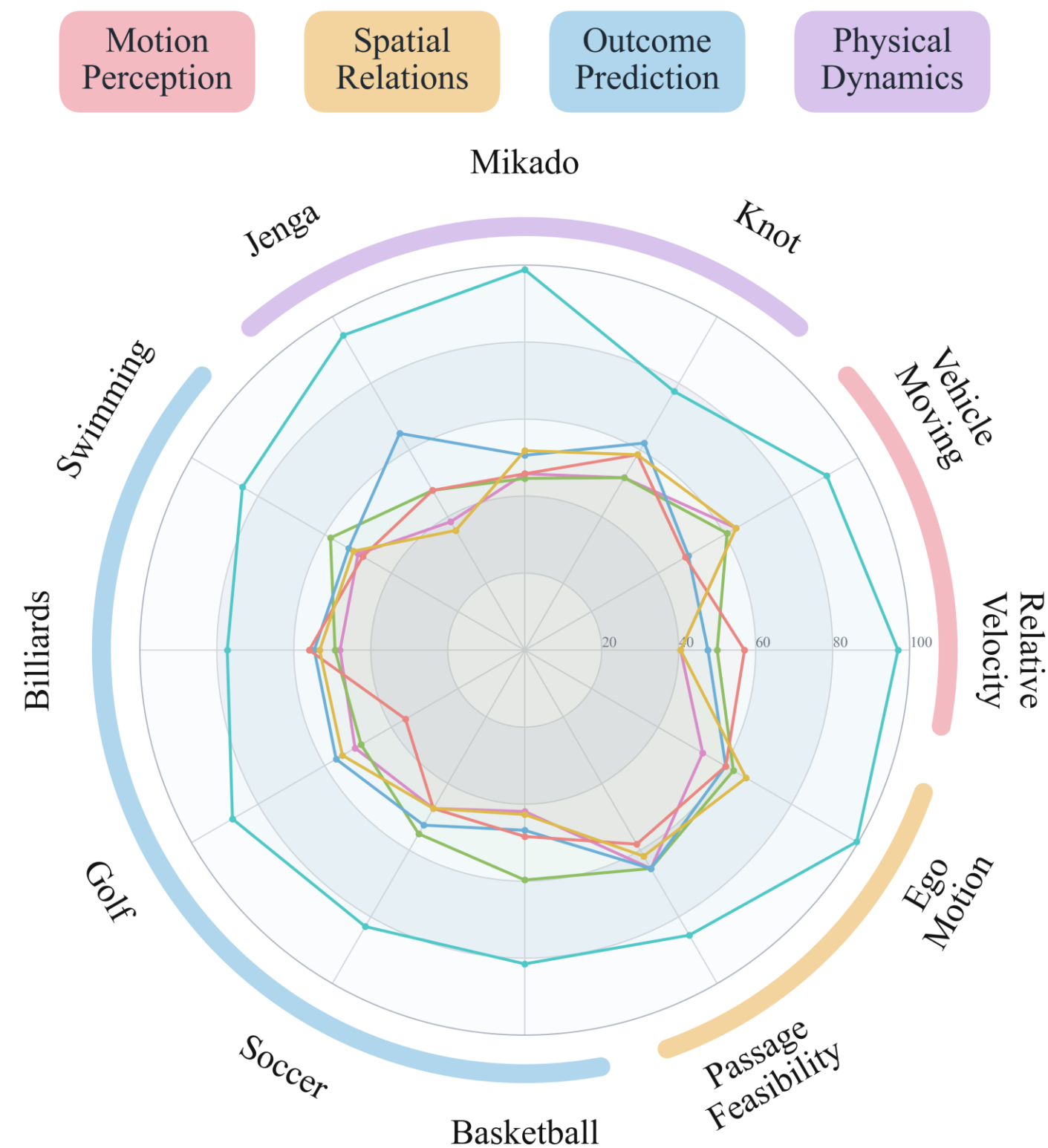


Proprietary General MLLMs



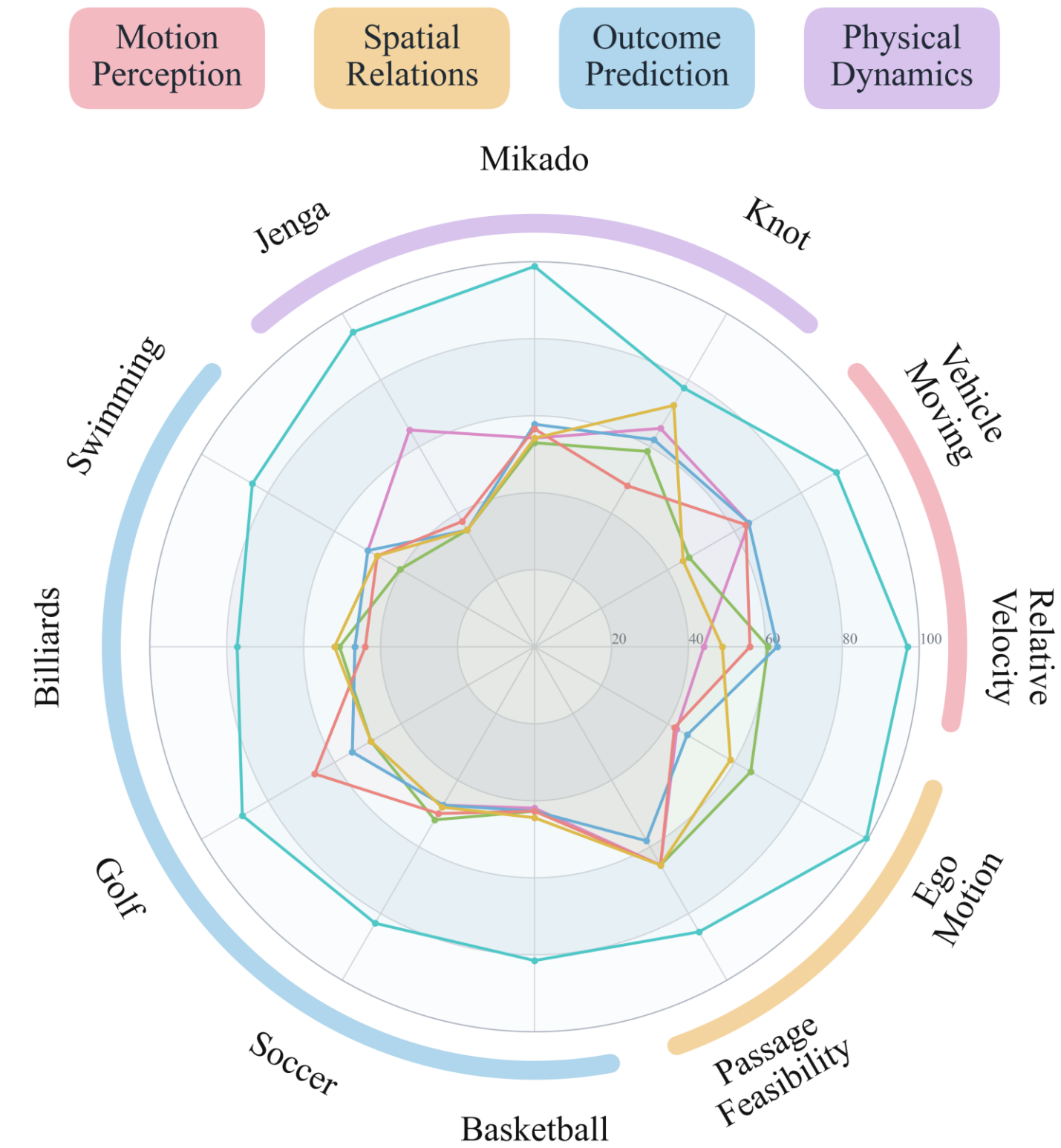
GPT-5.4-thinking
 Gemini-3.1-Pro-Preview
 Claude-Sonnet-4.6-thinking
 Claude-Opus-4.6-thinking
 Seed-2.0-Pro-thinking
 Human

Open-source General MLLMs



LLaVA-OneVision-1.5-8B
 Qwen3.5-27B-thinking
 InternVL3.5-38B
 Intern-S1-thinking
 GLM-4.6V-thinking
 Human

Specialized Spatial MLLMs



GeoThinker-Qwen2.5VL-7B
 SpaceR-SFT-7B
 Spatial-MLLM-820K-4B
 VST-7B-RL
 Spatial-SSRL-7B
 Human